

FreightQuote AI Platform

Milestone 2 Instructions — Full-Stack AI/ML Integration & Advanced Security Engine

0. Mission & Objective

In Milestone 1 you built the User Authentication module — JWT session handling, Streamlit UI, SQLite credentials, and Gmail-based OTP verification. Milestone 2 unifies that security gateway with the multi-agent ML core and the LLM Copilot, and adds three hardening layers: progressive account lockout, dynamic password strength verification, and a fully functional Admin Dashboard with Add / Delete / Unlock user lifecycle controls. Each of the three ML agents trains on 2 assigned Kaggle datasets (listed in Section 7.1) and must be expanded to compare 5+ algorithms before selecting a champion model.

1. Repository & Folder Structure

This project continues inside your existing Infosys Repository — do not create a separate GitHub repository for this milestone.

- Inside your existing Infosys Repository, create a new folder named Milestone2, alongside your Milestone1 folder.
- Everything for this milestone — README, notebook, modular scripts, screenshots — lives inside that folder.
- Final notebook file name: FreightQuote_AI_Milestone2.ipynb
- Modular Python scripts alongside it: auth.py, db.py, ui_theme.py, admin_dash.py, train_ml_freight.py, llm_engine_freight.py, plus requirements.txt
- Add a screenshots/ sub-folder inside Milestone2 for the captures listed in Section 9.

2. README.md (inside Milestone2)

Write, in your own words, what Milestone 2 adds on top of Milestone 1, which features were built, the tech stack used, and how to run the notebook. Also include:

- A clear system architecture overview (see Section 4 phase table).
- A localized Indian port coverage table: Mumbai (JNPT), Mundra, Chennai, Cochin.
- Step-by-step instructions for Colab Secrets and Kaggle API setup (Sections 3–4).
- A Screenshots section at the bottom, linking every image captured in Section 9.

3. Colab Runtime, GPU & Secrets Setup

3.1 Switch the Runtime to GPU

Milestone 2 loads Qwen2.5-3B-Instruct (4-bit quantized via bitsandbytes) to power the AI Copilot page. The 4-bit 3B model has a much smaller memory footprint than the full-precision 7B model and loads comfortably on a single T4.

- Runtime → Change runtime type → T4 GPU → Save.
- Run `!nvidia-smi` as the first code cell to confirm the GPU is attached before continuing.
- Load the model with `load_in_4bit=True` (bitsandbytes) to keep VRAM usage low and load time fast.

3.2 Create a Kaggle API Token (Recommended)

This lets the pricing/route/compliance models train on real logistics data (SCMS Delivery + DataCo Supply Chain) instead of synthetic data. The notebook must still work without it.

- Log in at kaggle.com → profile picture → Settings → API section → Create New Token.
- This downloads a `kaggle.json` containing a username and key.
- Add both as Colab Secrets (Section 3.3), or upload the file to `~/kaggle/kaggle.json`.

3.3 Store All Secrets in Colab (never hard-code them)

Click the key icon (Secrets) in the left sidebar, add each secret below, and toggle notebook access ON for each.

Secret Name	How to Get It	Used For
JWT_SECRET_KEY	Any long random string / passphrase you make up — never transmitted, only signs tokens locally	Signs & verifies login session tokens
ADMIN_EMAIL_ID	Any email you choose — becomes the Admin Panel login (seed: <code>infosys@ai</code> as fallback default)	Bootstraps the admin account on first run
ADMIN_PASSWORD	Any password meeting the strength rule (8+ chars, upper, lower, number, symbol)	Bootstraps the admin account on first run
NGROK_AUTHTOKEN	Free account at ngrok.com → dashboard → copy Authtoken	Gives the Streamlit app a public HTTPS URL
HF_TOKEN	HuggingFace account → Settings → Access Tokens	Authenticates HuggingFace LLM Copilot inference (Qwen2.5-3B (4-bit))
EMAIL_ID	The Gmail address OTP/alert emails are sent from	Sender address for real OTP emails (optional — console fallback works without it)
EMAIL_PASSWORD	Gmail → 2-Step Verification → App Passwords → create a 16-character app password	Authenticates the Gmail SMTP sender for OTP emails
KAGGLE_USERNAME / KAGGLE_KEY	From the <code>kaggle.json</code> downloaded in 3.2	Optional — trains models on real Kaggle data instead of synthetic

4. System Architecture — 4 Phases

Phase	Module / Component	Responsibility & Workflow
Phase 1: Security Gateway	Authentication & JWT	Enforces Login, Registration, and Forgot Password (Gmail OTP) before unlocking the UI. Stores hashed credentials and progressive logout state in SQLite (<code>users</code> table).
Phase 2: Domain Intelligence	3 Autonomous Agents	Once authenticated, unlocks Agent 1: Dynamic Pricing, Agent 2: Route Delay Classifier, and Agent 3: Carrier Compliance Sentinel tabs.
Phase 3: Generative Advisory	LLM Copilot & JSON	Integrates HuggingFace LLM orchestration (<code>llm_engine_freight.py</code>) to synthesize the 3 agents' numerical outputs into executive shipping strategies and structured JSON audit actions.
Phase 4: System Administration	Admin Dashboard	Dedicated administrative controls (<code>admin_dash.py</code>) restricted exclusively to users authenticated with role = 'Admin'.

5. Progressive Account Lockout (auth.py)

Extend the SQLite users table with failed_attempts INTEGER DEFAULT 0, lock_until TIMESTAMP DEFAULT NULL, and account_status TEXT DEFAULT 'active'. Enforce on every failed login attempt:

Failed Attempts	Action	User-Facing Message
3rd consecutive	lock_until = now() + 5 min (300s)	 Account temporarily locked for 5 minutes due to 3 failed attempts.
4th consecutive	lock_until = now() + 15 min (900s)	 Account temporarily locked for 15 minutes due to 4 failed attempts.
5th consecutive	account_status = 'locked', lock_until = NULL	 Account permanently locked due to 5 failed attempts. Only the System Administrator can unlock this account via the Admin Dashboard.

On a successful login where now() >= lock_until, reset failed_attempts = 0 and lock_until = NULL.

5.1 OTP Resend Rate Limiting (Gmail OTP Cooldown, tab3)

Track via session state or SQLite (otp_resend_count, otp_next_allowed):

Resend Attempt	Cooldown	Notification
1st resend	60 seconds	 Please wait 60 seconds before requesting another OTP.
2nd resend	180 seconds (3 min)	 Please wait 3 minutes before requesting another OTP.
3rd resend	300 seconds (5 min)	 Please wait 5 minutes before requesting another OTP.
4th+ resend	3,600 seconds (1 hour)	 Too many OTP requests. Please wait 1 hour before trying again.

6. Password Strength Policy & Real-Time Checker

Applies during registration (tab2) and password reset (tab3):

Length	Badge	Submission	Behavior
< 5 characters	 Weak	BLOCKED	st.warning('Password too weak (minimum 5 characters required).') — prevent database insertion.
5–9 characters	 Average	ALLOWED	 Average strength (10+ characters recommended for enterprise security).
10+ characters	 Good	ALLOWED	 Good password strength — proceed with bcrypt hashing.

7. ML Model Training — 5+ Algorithms per Agent

Each agent currently evaluates 3–4 algorithms across its 2 Kaggle datasets (SCMS Delivery + DataCo Supply Chain). Add at least 2 more distinct algorithms per agent in the training loop (train_ml_freight.py) so every agent compares 5+ models before saving the champion to joblib and logging metrics to the ml_models table.

Agent	Metric	Algorithms (5+)
Agent 1: Dynamic Pricing (Regression)	$R^2 \geq 0.90$	RandomForestRegressor, GradientBoostingRegressor, ExtraTreesRegressor, Ridge Regression, [NEW]

Agent	Metric	Algorithms (5+)
		DecisionTreeRegressor / AdaBoostRegressor / KNeighborsRegressor
Agent 2: Route Delay (Classification)	ROC-AUC optimization	RandomForestClassifier, GradientBoostingClassifier, LogisticRegression, SVC (RBF), [NEW] ExtraTreesClassifier / AdaBoostClassifier / KNeighborsClassifier
Agent 3: Carrier Audit (Classification)	ROC-AUC optimization	GradientBoostingClassifier, RandomForestClassifier, ExtraTreesClassifier, LogisticRegression, [NEW] DecisionTreeClassifier / AdaBoostClassifier / MLPClassifier

Confirm the printed R^2 for Agent 1 is ≥ 0.90 before moving on — the synthetic data generator is seeded, so a low score is unusual, but re-running the cell never hurts.

7.1 Kaggle Datasets per Agent

Pull each dataset with `kagglehub.dataset_download` using the slugs below, and save/rename the target file as listed for use in `train_ml_freight.py`.

Predictive Agent	Dataset Name / Role	Kaggle Dataset Slug	Target File Name
Agent 1: Pricing & Freight Cost Analyzer	SCMS Delivery Pricing Data	apoorvatsky/supply-chain-shipment-pricing-data	SCMS_Delivery_History_Dataset.csv
Agent 1: Pricing & Freight Cost Analyzer	DataCo Smart Supply Chain	shashwatwork/dataco-smart-supply-chain-for-big-data-analysis	DataCoSupplyChainDataset.csv
Agent 2: Route Delay & Risk Classifier	Supply Chain Analysis Data	harshsingh2209/supply-chain-analysis	supply_chain_data.csv
Agent 2: Route Delay & Risk Classifier	International Trade Logistics	victorchen/international-trade-logistics-dataset	trade_logistics.csv
Agent 3: Carrier Compliance Sentinel	Freight Carrier Performance	davidcariboo/freight-carrier-performance	carrier_perf.csv
Agent 3: Carrier Compliance Sentinel	Logistics Shipment Audit Data	suraj520/logistics-shipment-audit-data	audit_data.csv

8. AI Copilot Verification

Once the app launches (Section 10), open the AI Copilot page and submit a prompt, e.g. "Explain in 2 sentences why port congestion increases freight risk."

- If a GPU is attached and Qwen2.5-3B (4-bit) loaded successfully, you get a real model response.
- Otherwise you'll see a rule-based fallback — expected behavior, not a bug — but for the Milestone 2 screenshot you want the real model response, so confirm the GPU (Section 3.1) before capturing it.
- The Copilot must also synthesize Agent 1–3 outputs into a structured JSON audit action, per Phase 3 of Section 4.

9. Admin Dashboard — User Lifecycle Management

`admin_dash.py` already exists but is conditionally hidden behind `is_admin = (user_role.lower() == 'admin')`. Enable it for administrator accounts and add full lifecycle controls:

Feature	Scope	Implementation
Enable Admin Menu	Role verification	Seed the default account (Administrator, email: <code>infosys@ai</code> , pw: <code></code>)

Feature	Scope	Implementation
		admin@123 — or your ADMIN_EMAIL_ID/ADMIN_PASSWORD secrets) with role = 'Admin'.  Admin Dashboard appears in the sidebar when logged in as Admin.
Add User Portal	Admin creation form	st.form('add_user_form') letting the Admin create accounts with custom username, email, initial password, and role (Admin, Logistics Manager, etc.).
Delete User	Account revocation	 Delete button beside every user row; executes DELETE FROM users WHERE id=? on confirmation.
Unlock Account	Undo lockouts	For any user with account_status='locked' or failed_attempts>=3, show a  Unlock Account button. On click: UPDATE users SET failed_attempts=0, lock_until=NULL, account_status='active' WHERE id=?, then  User account unlocked successfully.
ML Model Card tab	Training transparency	Shows each agent's saved metrics (R ² /RMSE for Pricing, ROC-AUC for Route Delay & Carrier Audit) pulled from the ml_models table.

10. Launch, Verify & Finalize

10.1 Launch

Run the launch cell (Streamlit + ngrok tunnel). Open the printed public HTTPS URL and confirm:

- Login works with ADMIN_EMAIL_ID / ADMIN_PASSWORD.
- Home page shows a KPI overview.
- AI Copilot page returns a response (Section 8).
- ML Pricing Calculator returns a predicted cost for a sample distance/weight/congestion input.
- Admin Panel → ML Model Card tab shows R²/RMSE and the other agents' metrics.
- Progressive logout, OTP cooldown, and password strength badges all behave as specified in Sections 5–6.

10.2 Finalize, Clean Up & Upload

- Restart and re-run the notebook top to bottom to confirm it works cleanly.
- Clear all cell outputs (Edit → Clear all outputs) so no tokens or ngrok URLs remain visible.
- Search the notebook for any hard-coded email address, JWT secret, ngrok token, Kaggle key, or admin password — remove anything that slipped in, keeping only the Colab-secrets lookups.
- Download the notebook as .ipynb and upload it into the Milestone2 folder inside the Infosys Repository.

11. Screenshots to Capture

- Home page
- AI Copilot (prompt + response)
- ML Pricing Calculator (input + predicted cost)
- Admin Panel → ML Model Card tab (R²/RMSE and all agent metrics)
- Admin Panel → Add / Delete / Unlock user actions
- A triggered logout message and an OTP cooldown message

Save them in a screenshots/ folder inside Milestone2 and reference each one from README.md.

12. Final Checklist

- ☐ Infosys Repository → Milestone2 folder created
- ☐ README.md added, explains what Milestone 2 adds, includes port coverage table
- ☐ Colab runtime set to T4 GPU
- ☐ Kaggle API configured (or confirmed synthetic fallback works)
- ☐ All secrets (JWT, admin login, ngrok, HF_TOKEN, and optionally email + Kaggle) set up via Colab Secrets
- ☐ Login / Signup / Forgot Password (OTP) gate all agent & analytics tabs
- ☐ Progressive logout (5/15/permanent) implemented and tested
- ☐ OTP resend cooldown (60s / 180s / 300s / 1hr) implemented and tested
- ☐ Password strength checker (Weak/Average/Good) implemented on tab2 and tab3
- ☐ Agent 1 (Pricing): 5+ algorithms compared, $R^2 \geq 0.90$ confirmed
- ☐ Agent 2 (Route Delay): 5+ algorithms compared, ROC-AUC optimized
- ☐ Agent 3 (Carrier Compliance): 5+ algorithms compared, ROC-AUC optimized
- ☐ AI Copilot returns a real Qwen2.5-3B (4-bit) response and structured JSON audit action
- ☐ Admin Dashboard: Add User, Delete User, and Unlock Account all functional
- ☐ Admin Panel → ML Model Card tab shows metrics for all 3 agents
- ☐ Screenshots captured and linked in README.md
- ☐ All personal secrets removed from the notebook before upload
- ☐ Notebook restarted, re-run top to bottom, outputs cleared, and uploaded as FreightQuote_AI_Milestone2.ipynb